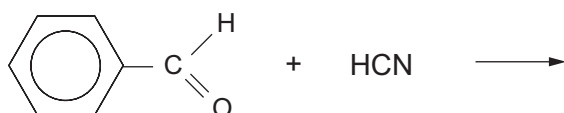


SECTION A

Answer all questions in the spaces provided.

1. (a) Complete the equation below, which shows the reaction of benzaldehyde with hydrogen cyanide. [1]



- (b) Give the **formula** of the nucleophile that takes part in the reaction shown in part (a). [1]

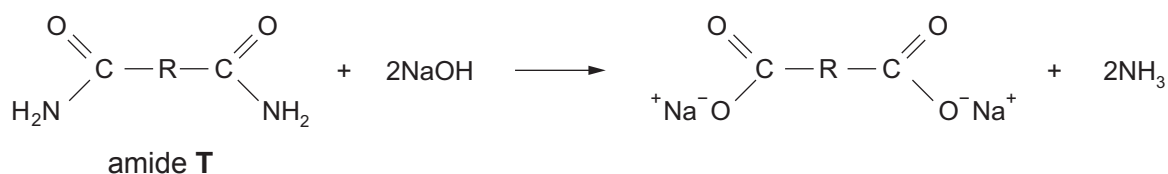
.....

2. Decarboxylation occurs when a carboxylic acid is heated with sodalime.

State the name of the carboxylic acid that will produce propane when strongly heated with sodalime. [1]

.....

3. Ammonia is produced when an amide is heated with aqueous sodium hydroxide.



In an experiment 3.90 g of the aliphatic amide **T** gave 0.060 mol of ammonia.

Use this information to find the formula of the hydrocarbon group R, present in the amide **T**. [2]

Formula

